Tyruko(natalizumab-sztn)

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1. Rationale for choice of product

- Tyruko (natalizumab-sztn) is ***the only biosimilar for the treatment of relapsing-remitting multiple sclerosis (RRMS) and Crohn's disease (CD) approved by the FDA and EMA***.[1]
- In 2025, Tyruko ***holds 18% of the European natalizumab market*** worth $500–700M, while *reducing treatment costs by up to 40% compared to its reference,* and is widely used in 14 European countries and the USA.[2]
- Tyruko is a **relevant case study in the emerging but highly profitable biosimilar drugs market**, *valued at 34.78 billion USD in 2025,* with an e*xpected value of 70.79 billion USD in 2036 globally.*[3]

2. Company profiles

Athena Neurosciences	***Athena Neurosciences:*** American biotechnology company that pioneered the early discovery and humanization of anti-VLA-4 monoclonal antibodies.
Elan	***Elan:*** Irish biotech company specializing in the development of therapies for neurodegenerative and autoimmune diseases.
Biogen	***Biogen:*** Multinational American company specialized in treatment of neurological diseases.
Polpharma Biologics	***Polpharma Biologics***: Poland-based company specialized in the development of biosimilar pharmaceuticals.
Sandoz	***Sandoz:*** Swiss global pharmaceutical company specialising in manufacturing and commercialization of generic and biosimilar pharmaceuticals.
Rezonbio	***Rezon Bio:*** Polish biologics contract development and manufacturing organisation (CDMO) brand launched by Polpharma Biologics in 2025.

3. Product Overview



- **ACTIVE SUBSTANCE:** Natalizumab (20 mg/mL) - a recombinant humanised IgG4κ[4]
- **REFERENCE**: Tysabri (Natalizumab), manufactured by Biogen.[5]
- **CHEMICAL PROPERTIES:** 146 kDa size; Asn300 N-linked glycosylation site on each heavy chain[4]
- **BIOSIMILARITY:** The high similarity demonstrated through comparative analytical and functional studies, supported by the Phase III ANTELOPE trial.[4]

Tyruko's Timeline

1992 — The scientific foundation for natalizumab development
Demonstration that antibodies against α4β1 integrin could inhibit lymphocyte binding to inflamed brain endothelium and reduce disease severity in an animal model of multiple sclerosis. [6]

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A monoclonal antibody targeting α4β1 integrin was first approved by the FDA for multiple sclerosis.[10]

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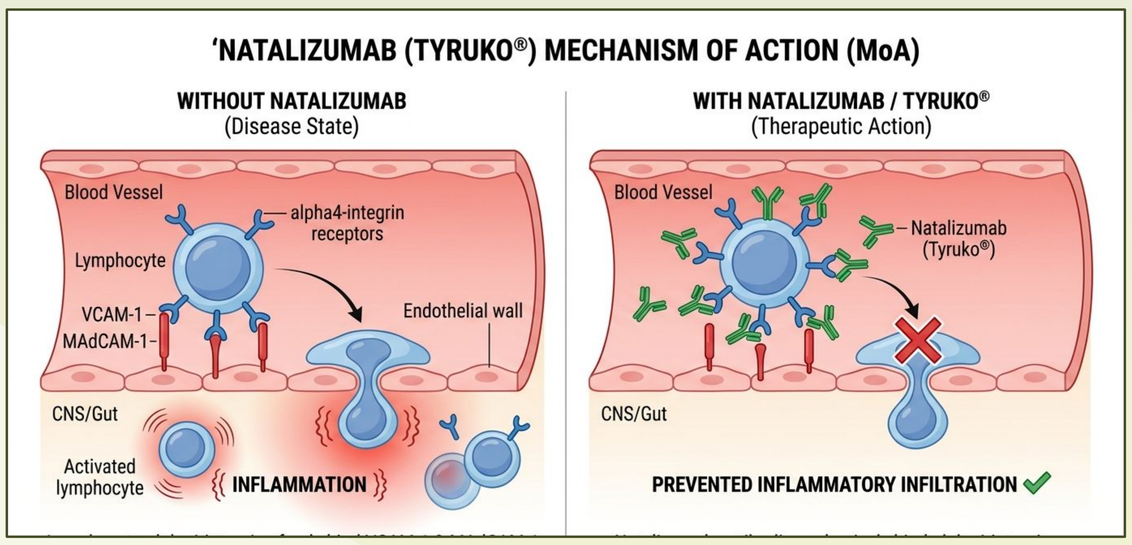
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Polpharma Biologics' CDMO business launched under the Rezon Bio brand, continuing the manufacturing of drug substance of Tyruko.[14]

Scientific development · Patent · Commercial agreements · Regulatory approvals

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4. Mechanism of action and medical indication



Medical indications[5]

- ***Relapsing-remitting multiple sclerosis (RRMS):*** Indicated as a *disease-modifying therapy (DMT)* for adults with highly active variants of RRMS.
- ***Crohn's disease (CD):*** Indicated for *inducing and maintaining clinical response* in adult CD patients with inadequate response/intolerance to conventional CD therapies and TNF-α inhibitors.

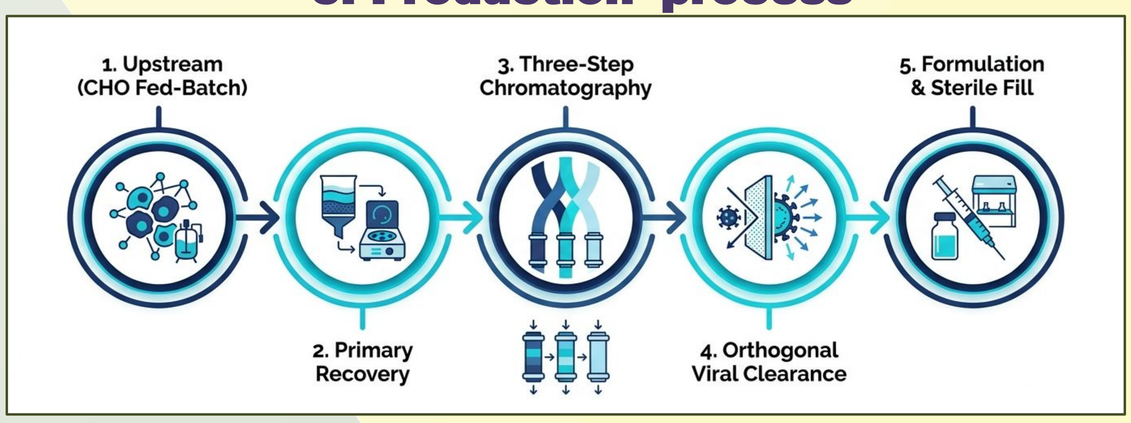
Biological mechanism of actions[6,15]

- ***Molecular target***
  Natalizumab utilises a selective adhesion-molecule (SAM) inhibition mechanism by binding to the α4-subunit of α4β1 and α4β7 integrins present on the lymphocyte surface.
- ***Target's role in pathogenesis***
  Targeted α4β1 and α4β7 integrins (SAM) bind VCAM-1 and MAdCAM-1 molecules presented on endothelium, enabling lymphocyte adhesion and subsequent entry into the parenchymal space and further tissue damage.
  Application of Natalizumab blocks lymphocyte migration from vessels across the Blood-Brain Barrier to the central nervous system in RRMS therapy and migration to gut mucosa in CD therapy.

Risk Evaluation and Mitigation Strategies (REMS)[5]

- ***PML Warning:*** The treatment is associated with a risk of life-threatening progressive multifocal leukoencephalopathy (PML), caused by opportunistic infection with John Cunningham virus (JCV).
- ***Mitigation strategy:*** Routine anti-JCV antibody screening assay is recommended for patients.

5. Production process



Tyruko's production process follows the general principles for recombinant immunoglobulin manufacturing. It can be divided into cultivation of recombinant cells producing IG (upstream processing) and purification of the obtained material to reach a pharmaceutical-grade product (downstream processing).[4]

Upstream processing

- **Cell Line:** Tyruko's manufacturing process uses Chinese Hamster Ovary (CHO) cells, replacing the murine NS0 cell line used for reference Tysabri.[4,17]
- **Product quality advantage:** Use of CHO cells reduces non-glycosylated heavy-chain variants and eliminates immunogenic murine Gal(α1–3)Gal motifs, providing higher glycosylation consistency compared to NS0 cells.
- **Fed-Batch Cultivation:** Antibody production is conducted in fed-batch mode using chemically defined, animal-component-free media certified as BSE/TSE-free.[4] Production volume is unknown, but it can be assumed to be single or multiple 2000L bioreactors, based on marketing data.

Downstream processing

- **Primary recovery:** production culture undergoes harvesting, during which cells are separated from the medium to clarify the material for further DSP processing.[4]
- **Multi-Step Purification:** consists of three different chromatography procedures that are disclosed in the regulatory reports.[4,16] The Tysabri reference process consists of a Protein A column, Anion Exchange Chromatography, and Hydrophobic Interaction Chromatography, which separates antibodies from host-cell proteins, DNA, and other impurities, captures Natalizumab molecules, and removes unwanted variants and aggregates.[4,17]
- **Orthogonal Viral Clearance:** To ensure compliance with EMA/ICH Q5A guidelines, removal of viral particles from purified antibodies is held in a dual orthogonal manner: by applying low-pH chemical inactivation (targeting enveloped viruses) and size-exclusion nanofiltration (targeting non-enveloped and enveloped viruses).[4,19]

Formulation

- **Histidine Buffer Composition:** Formulated as a 20 mg/mL concentrate (300 mg / 15 mL) in L-histidine, L-histidine HCl monohydrate, NaCl, and polysorbate 80 at pH 5.7.[4]
- **Patent advantage:** Replacing Tysabri's sodium phosphate buffer (pH 6.1) with its own buffers allowed faster commercialization of the product by omitting Biogen patents.[18]
- **Non-Terminal Sterilization:** Due to the nature of antibodies, the final product cannot be sterilized to avoid denaturation of the drug substance; thus, it undergoes 0,2µm filtration and aseptic filling.[4]

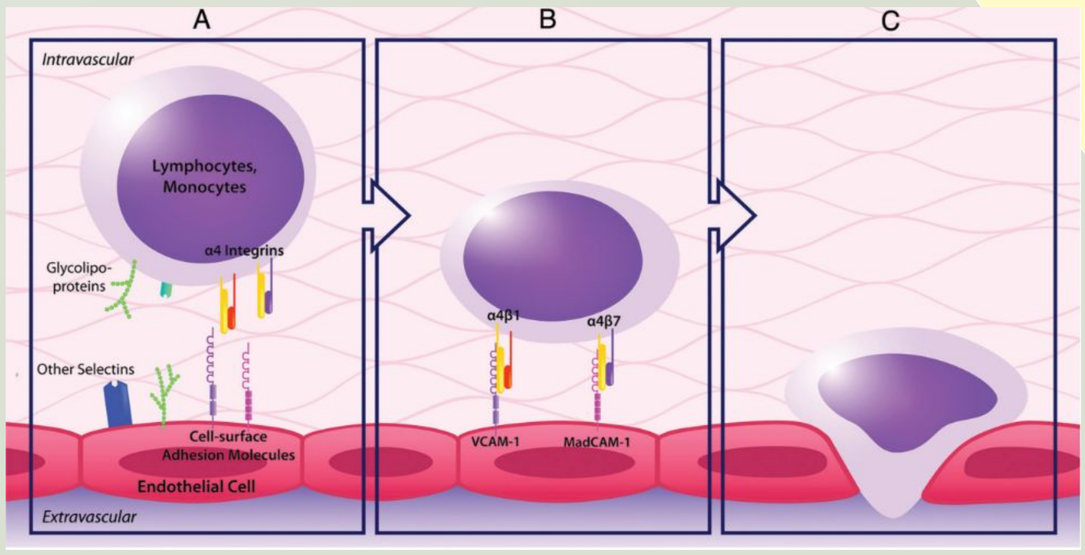
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Biogen and Elan collaborated in the development and commercialisation of natalizumab, originally known as Antegren and later marketed as Tysabri. Their collaboration was announced in 2000.
Rezon Bio is a Poland-based contract development and manufacturing organisation (CDMO) specialising in biologic medicines. Tyruko (natalizumab-sztn) was developed by scientists at Polpharma Biologics, which retained responsibility for manufacturing and supplying the active pharmaceutical ingredient. In 2025, Rezon Bio was launched as an independent CDMO built on Polpharma Biologics' biologics expertise and manufacturing infrastructure.

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Naturally leucocyte escape instravascular space and enters parenchymal space by rolling, adhesion , and extravasation mechanism. Natalizumab disables adhesion step for lymphocytes by binding α4β1 and α4β7 integrins molecules present on the leucocyte surface, blocking the access to integrins ligands - VCAM-1 and MAdCAM-1 - that are expressed on in inflamed cerebrovascular and intestinal endothelium. By blocking the escape from intravascular space Natalizumab reduce the autommunie reaction on cells resulting in delay of dieses progression.

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Product Overview



- *Tyruko* is a biosimilar to *Tysabri*, containing *Natalizumab*, a recombinant humanised IgG4κ subclass monoclonal antibody with approximately 146 kDa size.
- Each heavy chain has one N-linked glycosylation site at Asn300.
- It displays a natural amount of microheterogeneity in terms of a different degree of glycosylation and post-translational modifications of amino acids.
- The high similarity to the reference product Tysabri was demonstrated through comparative analytical and functional studies, supported by the Phase III ANTELOPE trial.

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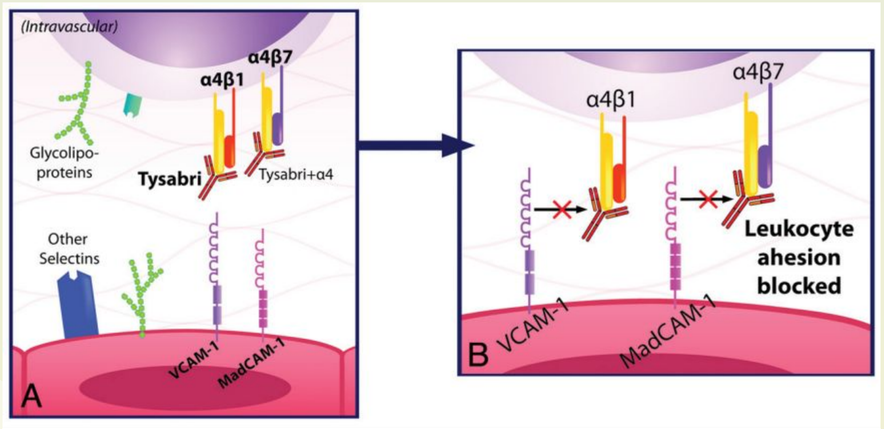
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Mechanism of action and medical indication



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Production process

Manufacturing process of Tyruko follows the general principles for recombinant antibo production, that can be dived into Upstream-processing and Downstream-processing.



Upstream-processing

The first part of the manufacturing process consisting of subsequent cultivation of recombinant cells in larger volumes to eventually start a production culture where the antibodies are produced.

Fig X (left): One of Rezon Bio's single use bioreactors used in manufacturing process of different products (Source:https://www.cphi-online.com/)

1. ***Cell banks:*** Unlike reference Tysabri, Tyruko uses disclosed CHO insted of NS0 cells. The production celline is stored in characterised WCB with genetic stability assessed by multiple studies. The single cell bank vial is the main starting material for the manufacturing process.
2. ***Cell propagation:*** Thawed WCB undergoes successive passages in different shake flasks until reaching sufficient cell density to inoculate the seed train bioreactor.
3. ***Inoculum expansion:*** cells cultivation in bioreactors using defined and animal product free media, to reach a density needed for inoculation of production bioreactor.
4. ***Production bioreactor:*** Registration documentation confirmed that the production is hold in fed-batch mode in defined animal-product free medium.



Downstream-processing

The second part of the manufacturing process removes all product impurities and undiresed proteins to eventually obtain a concentrated drug substance that meets high pharmaceutical standards.

Fig X (left): One of Rezon Bio's single use bioreactors used in manufacturing process of different products (Source:https://www.cphi-online.com/)

1. ***Primary recovery:*** This step aims to separate the cells from medium containing the target antibody.
2. ***Chromatography*:** The purpose of these steps is to separate antibodies from hosts proteins to eventually obtain a poll of desired variants of the target antibody. This procedure contains of three different chromatography procedures that are disclosed in the regulatory reports. Reference Tysabri process consist of Protein A column, Anion Exchange Chromatography and Hydrophobic Interaction Chromatography.
3. ***Virus filtration:*** In this step the antibody is separated from virus particles based on their sizes, to reduce risk of viral contamination in the final product.
4. ***Ultrafiltration/Diafiltration (UF/DF):*** Natalizumab is concentrated by ultrafiltration, while diafiltration exchanges the process buffer for the final formulation buffer. This step prepares the purified antibody for final formulation.
5. ***Sterile filtration and aseptic filling:*** Natalizumab is formulated in a sodium chloride, histidine, histidine hydrochloride monohydrate, polysorbate 80 and water for injections. Final product cannot be sterilised thus it undergoes sterile filtration and aseptic filling.

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Scientific development Patent Commercial agreements Regulatory approvals

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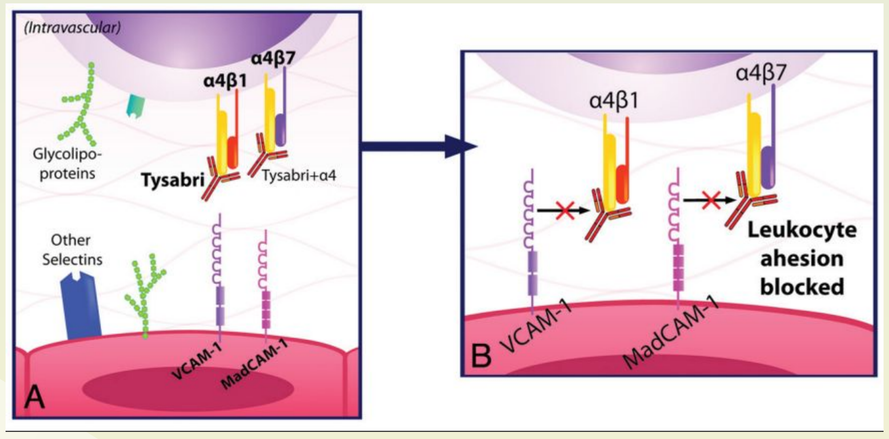
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Production process

Disclaimer: Detailed manufacturing instruction are proprietary information protected by companies involved in its production and commercialization, and cannot be found in publicly available sources. The process listed below based on data available from EMA report and patents concerning Natalizumab and Natalizumab-producing cell lines, and aims to provide a technical overview of manufacturing process rather than description of validated commercial process used for Tyruko manufacturing.

USP:

1. Cell banks: undetermined CHO celline stored in characterised WCB with genetic stability assessed by transgene sequence analysis, gene copy number comparison, and a Limit of In Vitro Cell Age (LIVCA) study.
2. Cell train: Thawed WCB undergoes successive passages in different shake flasks until reaching sufficient cell density to inoculate the seed train bioreactor.
3. Seed train: cells cultivation in bioreactors using defined and animal product free media, to reach a density needed for inoculation of production bioreactor.
4. Production bioreactor: Registration documentation confirmed that the production is hold in fed-batch mode in defined medium. Available patent indicated a possible usage of PolCHO-P CD medium, supplemented with citrulline, PolCHO-P Feed A, PolCHO-P Feed B and glucose. Reported experimental parameters include a pH setpoint 7.00, DO setpoint 40% ,temperature shift from 36,8°C to 33°C in day 5 and culture duration of 14 days. These conditions are patent-reported examples and should not be interpreted as confirmed commercial manufacturing parameters for Tyruko.

DSP:

1. Primary recovery: EMA report confirms that Tyruko undergoes primary recovery process, without specifying what kind of technique is used. The patents for reference Tysabri mentioned usage of 0,2 µm membrane filters before chromatography steps.
2. Chromatography - EMA report confirms that the proces includis several chromatography steps, without further details. Reference Tysabri undergoes three step purification suing Protein A column, Anion Exchange Chromatography and Hydrophobic Interaction Chromatography.
3. Virus filtration - In Tysabari process target protein s passed through a 15 nm nanofilter to remove potential viral contaminants. This physical virus-removal step complements low-pH viral inactivation and contributes to the viral safety of the product. For Tyruko there are no available details on the virus filtration
4. Ultrafiltration/Diafiltration (UF/DF): Natalizumab is concentrated by ultrafiltration, while diafiltration exchanges the process buffer for the final formulation buffer. This step prepares the purified antibody for final formulation.
5. Final formulation: Following UF/DF, natalizumab is formulated in a sodium chloride, histidine, histidine hydrochloride monohydrate, polysorbate 80 and water for injections. Final product cannot be sterilised thus it undergoes sterile filtration and aseptic filling.